



Stochastic Processes

Lecture 16

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Random Mapping Representation

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Random Mapping Representation)

Consider a TPM P on a discrete state space $\mathcal{X} \subset \mathbb{R}$.

A **random mapping representation** of P is a tuple (Z, f) where

- $Z : \Omega \rightarrow \Lambda$ is a random variable,
- $f : \mathcal{X} \times \Lambda \rightarrow \mathcal{X}$ is a measurable function,

satisfying the condition

$$\mathbb{P}\left(\left\{f(x, Z) = y\right\}\right) = P(x, y) \quad \forall x, y \in \mathcal{X}.$$

Random Mapping Representation

Theorem (Random Mapping Representation)

Given a TPM P on a discrete state space $\mathcal{X} \subset \mathbb{R}$, a random mapping representation always exists.

Proof of Theorem 1:

- The proof involves construction of $Z : \Omega \rightarrow \Lambda$ and $f : \mathcal{X} \times \Lambda \rightarrow \mathcal{X}$
- Let $\Lambda = (0, 1)$ and let $Z \sim \text{Uniform}(0, 1)$
- For each x , let $F_x(\cdot)$ denote the CDF corresponding to row x of P , i.e.,

$$F_x(y) = \sum_{x' \leq y} P(x, x'), \quad y \in \mathcal{X}.$$

- Let $f : \mathcal{X} \times (0, 1) \rightarrow \mathcal{X}$ be defined as

$$f(x, z) = \inf\{y \in \mathcal{X} : F_x(y) \geq z\}.$$

Random Mapping Representation

Theorem (Random Mapping Representation)

Given a TPM P on a discrete state space $\mathcal{X} \subset \mathbb{R}$, a random mapping representation always exists.

Proof of Theorem 1:

- Then, for all $x, y \in \mathcal{X}$, we have

$$\mathbb{P}\left(\left\{f(x, Z) \leq y\right\}\right) = F_x(y)$$

For a proof of this, see inverse CDF sampling technique ([Lecture 21, AI5030 \(Jul-Nov 2025\)](#))

- Hence, it follows that

$$f(x, Z) \sim P(x, \cdot) \quad \forall x \in \mathcal{X}.$$

Simulating a (Time) Homogeneous Markov Chain

Main Question

**Given (μ, P) , is there a process $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ such that $\mathbf{X} \stackrel{\text{MC}}{\sim} (\mu, P)$?
If so, how do we simulate it?**

- From Theorem 1, we know that there exists a mapping f corresponding to P
- Generate $Z_1, Z_2, \dots \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1)$
- Generate $X_1 \sim \mu$ independent of Z_1
- For each $n \geq 2$, set

$$X_n = f(X_{n-1}, Z_n).$$

- **Exercise:** The process $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ is a time homogeneous DTMC with $\mathbf{X} \stackrel{\text{MC}}{\sim} (\mu, P)$

Simulating a Non-Homogeneous Markov Chain

Main Question

Given a PMF μ and a family of stochastic matrices $\{P^{(n)} : n \in \mathbb{N}\}$ on a finite set $\mathcal{X}$, is there a Markov process $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ that fits $(\mu, \{P^{(n)}\}_{n \in \mathbb{N}})$? If so, how do we simulate it?

- From Theorem 1, we know that for each $n \in \mathbb{N}$, there exists a mapping f_n corresponding to $P^{(n)}$
- Generate $Z_1, Z_2, \dots \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1)$
- Generate $X_1 \sim \mu$ independent of Z_1
- For each $n \geq 2$, set

$$X_n = f_n(X_{n-1}, Z_n).$$

- **Exercise:** The process $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ is a DTMC with initial distribution μ and TPMs $\{P^{(n)} : n \in \mathbb{N}\}$

Strong Markov Property

Strong Markov Property

- Suppose that $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ is a homogeneous DTMC on a finite state space $\mathcal{X}$ and having TPM P
- **Markov property:** For any **deterministic** $n \in \mathbb{N}$ and $(x_1, \dots, x_{n+1}) \in \mathcal{X}^n$, we have

$$\mathbb{P}\left(\left\{X_{n+1} = x_{n+1}\right\} \mid \left\{X_n = x_n, \dots, X_1 = x_1\right\}\right) = \mathbb{P}\left(\left\{X_{n+1} = x_{n+1}\right\} \mid \left\{X_n = x_n\right\}\right).$$

Does the above Markov property hold when n is replaced with a random variable τ ?
YES, when τ is a stopping time! This is captured by the strong Markov property.

Strong Markov Property

Lemma (Strong Markov Property)

Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ be a homogeneous DTMC on a finite state space $\mathcal{X}$ with TPM P .

Let τ be a **stopping time** adapted to the natural filtration of the process $\mathbf{X}$. Then, for all $y \in \mathcal{X}$ and $n \in \mathbb{N}$,

$$\mathbb{P}\left(\left\{X_{\tau+n} = y\right\} \mid \tau, X_1, \dots, X_\tau\right) = \mathbb{P}\left(\left\{X_{\tau+n} = y\right\} \mid \tau, X_\tau\right).$$

Proof of Lemma 3

- For any $m \geq 2$ and $(x_1, \dots, x_m) \in \mathcal{X}^m$, we have

$$\begin{aligned}
 \mathbb{P}\left(\{X_{\tau+n} = y\} \mid \{\tau = m, X_1 = x_1, \dots, X_m = x_m\}\right) &= \mathbb{P}\left(\{X_{m+n} = y\} \mid \{\tau = m, X_1 = x_1, \dots, X_m = x_m\}\right) \\
 &= \mathbb{P}\left(\{X_{m+n} = y\} \mid \{X_1 = x_1, \dots, X_m = x_m\}\right) \\
 &= \mathbb{P}\left(\{X_{m+n} = y\} \mid \{X_m = x_m\}\right) \\
 &= \mathbb{P}\left(\{X_{m+n} = y\} \mid \{\tau = m, X_m = x_m\}\right) \\
 &= \mathbb{P}\left(\{X_{\tau+n} = y\} \mid \{\tau = m, X_m = x_m\}\right).
 \end{aligned}$$

- The second and penultimate equalities above follow from the fact that

$$\{\tau = m\} \in \sigma(X_1, \dots, X_m),$$

as τ is a stopping time

A Corollary to Lemma 3

- If $\mathbf{X} \stackrel{\text{MC}}{\sim} (\mu, P)$ and τ is a stopping time adapted to the natural filtration of $\mathbf{X}$, then

$$\mathbb{P}\left(\{X_{\tau+n} = y\} \mid \{X_{\tau} = x\}\right) = P^n(x, y) \quad \forall n \in \mathbb{N}, x, y \in \mathcal{X}.$$

Proof:

— We have

$$\begin{aligned} \mathbb{P}\left(\{X_{\tau+n} = y, X_{\tau} = x\}\right) &= \mathbb{P}\left(\{X_{\tau+n} = y, X_{\tau} = x, \tau < +\infty\}\right) \\ &= \sum_{m \geq 2} \mathbb{P}\left(\{X_{\tau+n} = y, X_{\tau} = x, \tau = m\}\right) \\ &= \sum_{m \geq 2} \mathbb{P}\left(\{X_{m+n} = y, X_m = x, \tau = m\}\right) \\ &= \sum_{m \geq 2} \mathbb{P}\left(\{X_m = x, \tau = m\}\right) \cdot \mathbb{P}\left(\{X_{m+n} = y\} \mid \{X_m = x, \tau = m\}\right) \\ &= \sum_{m \geq 2} \mathbb{P}\left(\{X_m = x, \tau = m\}\right) \cdot P^n(x, y) = \mathbb{P}\left(\{X_{\tau} = x\}\right) \cdot P^n(x, y). \end{aligned}$$

Hitting Times

Hitting Times

Definition (Hitting Time)

Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}} \stackrel{\text{MC}}{\sim} (\mu, P)$ be DTMC on a finite state space $\mathcal{X}$. Fix $y \in \mathcal{X}$.

Let $\tau_y^{(0)} := 1$, and let

$$\tau_y^{(k)} = \inf\{n > \tau_y^{(k-1)} : X_n = y\}, \quad k \in \mathbb{N}.$$

The random variable $\tau_y^{(k)}$ is called the **k th hitting time of state y** .

Note: At each hitting time $\tau_y^{(k)}$, we have

$$X_{\tau_y^{(k)}} = y.$$

Lemma (Hitting Time is Adapted)

For each $k \in \mathbb{N}$, the random variable $\tau_y^{(k)}$ is adapted to the process $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$, i.e.,

$$\{\tau_y^{(k)} = m\} \in \sigma(X_1, \dots, X_m) \quad \forall m \geq 2.$$

Proof of Lemma 4

- **Base case $k = 1$:** We note that

$$\{\tau_y^{(1)} = m\} = \{X_1 \neq y, \dots, X_{m-1} \neq y, X_m = y\} \in \sigma(X_1, \dots, X_m).$$

- Assuming that $\tau_y^{(k)}$ is a stopping time for some $k \geq 2$, we have

$$\{\tau_y^{(k+1)} = m\} = \bigcup_{\ell=1}^{m-1} \left\{ \tau_y^{(k)} = \ell, X_{\ell+1} \neq y, \dots, X_{m-1} \neq y, X_m = y \right\} \in \sigma(X_1, \dots, X_m)$$

Remark

If $\mathbb{P}(\{\tau_y^{(k)} < +\infty\}) = 1$, then $\tau_y^{(k)}$ is a stopping time.

Whether or not $\tau_y^{(k)}$ is a stopping time hinges strongly on the structure of (μ, P) .

Computing $\mathbb{P}(\{\tau_Y^{(k)} < +\infty\})$

- Considering the base case $k = 1$, we have

$$\begin{aligned}
 \mathbb{P}(\{\tau_Y^{(1)} < +\infty\}) &= \sum_{m \geq 2} \mathbb{P}(\{\tau_Y^{(1)} = m\}) \\
 &= \sum_{m \geq 2} \mathbb{P}(\{X_1 \neq \gamma, \dots, X_{m-1} \neq \gamma, X_m = \gamma\}) \\
 &= \sum_{m \geq 2} \sum_{x_1 \neq \gamma} \cdots \sum_{x_{m-1} \neq \gamma} \mathbb{P}(\{X_1 = x_1, \dots, X_{m-1} = x_{m-1}, X_m = \gamma\}) \\
 &= \sum_{m \geq 2} \sum_{x_1 \neq \gamma} \cdots \sum_{x_{m-1} \neq \gamma} \mu(x_1) \cdot \left(\prod_{t=2}^{m-1} P(x_{t-1}, x_t) \right) \cdot P(x_{m-1}, \gamma)
 \end{aligned}$$

An Important Observation Regarding $\tau_Y^{(k)}$

Lemma (An Important Observation Regarding $\tau_Y^{(k)}$)

Fix $k \in \mathbb{N}$ and $y \in \mathcal{X}$, and suppose that $\mathbb{P}(\{\tau_Y^{(k)} < +\infty\}) = 1$.

Then, the history up to $\tau_Y^{(k)}$ is conditionally independent of the future conditioned on $\tau_Y^{(k)}$, i.e.,

$$(\tau_Y^{(k)}, X_1, \dots, X_{\tau_Y^{(k)}}) \perp\!\!\!\perp X_{\tau_Y^{(k)}+n} \mid \tau_Y^{(k)} \quad \forall n \in \mathbb{N}.$$

Proof of Lemma:

- For any $m \geq 2$, $(x_1, \dots, x_{m-1}) \in \mathcal{X}^{m-1}$, and $z \in \mathcal{X}$, we have

$$\begin{aligned} & \mathbb{P}\left(\{X_{\tau_Y^{(k)}+n} = z\} \mid \{\tau_Y^{(k)} = m, X_1 = x_1, \dots, X_{m-1} = x_{m-1}, X_m = y\}\right) \\ &= \mathbb{P}\left(\{X_{m+n} = z\} \mid \{\tau_Y^{(k)} = m, X_1 = x_1, \dots, X_{m-1} = x_{m-1}, X_m = y\}\right) \\ &= \mathbb{P}\left(\{X_{m+n} = z\} \mid \{X_m = y\}\right) = \mathbb{P}\left(\{X_{m+n} = z\} \mid \{\tau_Y^{(k)} = m, X_m = y\}\right) \\ &= \mathbb{P}\left(\{X_{m+n} = z\} \mid \{\tau_Y^{(k)} = m\}\right) = \mathbb{P}\left(\{X_{\tau_Y^{(k)}+n} = z\} \mid \{\tau_Y^{(k)} = m\}\right). \end{aligned}$$

Increments of Hitting Times

- By definition, we have

$$1 = \tau_Y^{(0)} < \tau_Y^{(1)} < \tau_Y^{(2)} < \tau_Y^{(3)} < \dots$$

- If $\tau_Y^{(k)}$ is a **stopping time** for each k , then:

- **Exercise:** $\tau_Y^{(k+1)} - \tau_Y^{(k)}$ is adapted to the filtration

$$\mathcal{H} = \{\mathcal{H}_n : n \in \mathbb{N}\}, \quad \mathcal{H}_n := \mathcal{G}_{\tau_Y^{(k)} + n}.$$

- $\tau_Y^{(k+1)} - \tau_Y^{(k)}$ is finite with probability 1

$\tau_Y^{(k+1)} - \tau_Y^{(k)}$ is a **stopping time (adapted to a slightly different filtration)**

- One may then envision the following “blocks”:

$$\underbrace{\left(\tau_Y^{(1)}, X_1, \dots, X_{\tau_Y^{(1)}}\right)}_{\text{Block 1}}, \underbrace{\left(\tau_Y^{(2)} - \tau_Y^{(1)}, X_{\tau_Y^{(1)}+1}, \dots, X_{\tau_Y^{(2)}}\right)}_{\text{Block 2}}, \underbrace{\left(\tau_Y^{(3)} - \tau_Y^{(2)}, X_{\tau_Y^{(2)}+1}, \dots, X_{\tau_Y^{(3)}}\right)}_{\text{Block 3}}, \dots$$

- Each state $y \in \mathcal{X}$ may potentially have a different block structure